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BoneSE: Bone Conduction-Assisted Speech Enhancement Based on COTS Earphone

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***Abstract*—Speech enhancement is crucial for reliable communication in noisy environments. However, the lack of a priori knowledge about target speech characteristics in conventional systems often leads to erroneous extraction of interfering speech as desired signals during noise suppression, significantly compromising system performance. Recently, researchers have proposed to use the side-channel signal as an assist to denoise noisy speech. This paper proposes BoneSE, a multimodal speech enhancement approach using bone conduction. The basic idea of BoneSE is to perceive the bone-conducted sound with an IMU sensor embedded in the Commercial Off-The-Shelf (COTS) earphone and then leverage the correlations between the bone conduction signal and audio signal for speech enhancement. However, the lack of high-frequency components in the IMU modality brings data imbalance and hinders data fusion. To address this challenge, we explore and model the relationship between multimodal signals and design a fusion module according to the time and frequency correlation. Moreover, to balance fast processing and denoising performance, we propose bone conduction-based noise level metrics to measure the noise level. To accommodate different noise levels, we propose an adaptive model selection approach based on reinforcement learning to select the proper denoising model, thereby optimizing the latency. Experiments on two datasets show that the proposed method performs favorably against state-of-the-art methods and can enhance speech in low SNR scenarios.**

***Index Terms*—Speech enhancement, Bone conduction, Earphone, Multimodal fusion.**

I. Introduction

MAKING phone calls and sending voice messages in noisy environments have always been a headache. For example, when answering the phone in the subway or on a crowded street, the speaker's voice is often drowned out by ambient noises, which seriously interferes with the quality of the call and makes it difficult for the remote listener to clearly understand the speech content. To capture clear speech in noisy environments, researchers propose to process the audio signal digitally to increase the signal-to-noise ratio (SNR), i.e., enhance the speech and suppress background noise.

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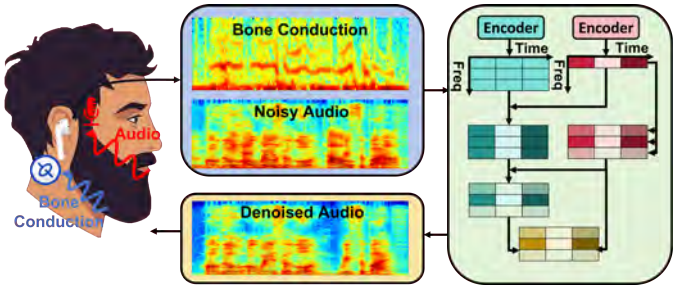

Fig. 1: Speech enhancement based on bone conduction.

In recent years, there has been an increasing interest in speech enhancement with the development of deep learning. According to the signal modes used, related works can be classified into audio-based approaches and multimodal approaches. Audio-based approaches [1]–[3] conduct speech enhancement with only audio signals. Researchers utilize well-designed artificial networks to separate target speeches from noisy audio by analyzing the internal structure of the user's voice. However, audio-based approaches have no prior knowledge of the target audio, which may lead to the error enhancement of external interfering speech when the target speech is relatively weak. Multimodal approaches utilize side-channel signals, such as mmWave [4], ultrasonic [5], [6], and video [7], to assist speech enhancement. For example, researchers [5] utilize ultrasonic to sense mouth movement and leverage the correlation between the side-channel signal and audio signal for speech enhancement. However, choosing an effective side-channel that can benefit speech enhancement is very challenging.

This paper proposes BoneSE, a bone conduction-assisted speech enhancement approach using earphones. The idea of BoneSE is based on our finding that the bone conduction signal is an effective side-channel signal for speech enhancement. Specifically, when the human speaks, parts of the voice can penetrate the vocal tract and reach the auditory ossicle via bone conduction. The bone conduction can resist the distortion of external noise and has a strong correlation with the air-conducted sound. These features are suitable to assist speech enhancement. As shown in Fig. 1, when the user wears an earphone, the embedded microphone collects the audio and the embedded Inertial Measurement Unit (IMU) perceives the bone conduction signal. Then BoneSE would leverage the time and frequency correlations between multimodal signals for

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speech enhancement.

BoneSE has two advanced features. 1) Most Commercial Off-The-Shelf (COTS) earphones have microphones and IMU sensors embedded in them. Therefore, users only need to wear earphones for speech enhancement without extra devices or advanced deployment. 2) Bone conduction sound is perceived by contact sensing and will not be disturbed by ambient noise, which allows bone conduction signal to be a reliable reference for separating clean speech even in highly noisy environments.

We address two critical challenges in BoneSE. *The first challenge is how to use the low-frequency bone conduction signal to enhance human speech across a broadband frequency range.* Actually, only the low-frequency bone conduction signal can be captured by the IMU sensor, while human voice covers a frequency range of 80Hz to 8KHz. Therefore, there exists a huge gap between the obtained bone conduction signal and the target speech, making it challenging to enhance broadband human voice with the low-frequency signal. To address this challenge, we deeply explore the correlation between the bone conduction signal and the audio signal. We find that the high-frequency components of the audio signal are highly correlated with the low-frequency bone conduction signal. Therefore, we propose an artificial network called BoneSEnet based on the attention mechanism to extract the target speech. BoneSEnet utilizes the channel and time attention block (CTA) to allocate attention according to the time synchronization cross-modally and utilizes the frequency attention block (FA) to allocate attention leveraging the frequency correlation of the multimodal signals. *The second challenge is how to speed up speech enhancement with limited computing resources.* Using deep neural networks for speech enhancement inevitably introduces a long processing time. To address this challenge, we employ the bone conduction signal to determine the necessity of time-consuming processes and optimize the trade-off between noise suppression efficacy and computational latency. Specifically, we design a bone conduction-based voice activity detector to detect audio segments containing the target user's speech for processing. Moreover, to achieve fast processing while ensuring denoising performance, we choose the appropriate denoising model for different noise levels. We propose bone conduction-based noise level metrics to measure the noise level. To accommodate different noise levels, we propose an adaptive model selection approach based on policy gradient to adaptively select denoising models with different computational amounts. In this way, we can speed up the speech enhancement without compromising the performance of noise reduction.

While there have been some attempts to use bone conduction for speech enhancement, our work presents completely different innovations and contributions. Our contributions to this work are threefold. Firstly, we delve deeply into the correlation between the bone conduction signal and human speech to model their relationship. Secondly, we propose an attention-based network called BoneSEnet, innovatively leveraging the correlation between multiple modalities to guide attention allocation for data fusion, and propose an adaptive model selection approach to achieve a trade-off between denoising performance and time cost, optimizing the overall denoising process. Finally, we implement a prototype of BoneSE and conduct comprehensive experiments for evaluation. Experimental results show that BoneSE outperforms the state-of-the-art speech enhancement approaches and can enhance speech in low SNR scenarios such as subway and bar.

II. Related Work

A. Audio-based speech enhancement

Audio-based approaches enhance speech with only noisy speech signals, they can be further classified into single-channel approaches and multi-channel approaches. Single-channel approaches denoise speech by analyzing noisy speech signals in the time domain [8]–[13] or time-frequency domain [1]–[3], [14]–[19]. For example, SEGAN [8] designs a generative adversarial network to extract target speech in the time domain. PHASEN [18] learns the harmonic information in the time-frequency domain and designs a two-stream architecture to predict amplitude masks and phase masks separately for the speech extraction. *However, single-channel approaches may extract external interfering speech when multiple speeches are mixed.* Multi-channel approaches [20]–[22] take noisy speech signals from multiple microphones as inputs for speech enhancement. For example, ClearBuds [22] designs an artificial network to perform beamforming with audio signals of two microphones. It can extract target speech by analyzing the differences between multi-channel audio signals. *However, multi-channel approaches are susceptible to interference from noise sources near the user.*

B. Multimodal Speech Enhancement

Multimodal approaches leverage the correlation between multimodal signals and the audio signal to assist speech enhancement. Researchers utilize signal modes including video [23]–[28], mm-wave [4], [29]–[31], ultrasonic [5], [6], [32], vibration [7], [33]–[36] for noise reduction. For example, Pan et.al. [23] propose to extract features from the video of lip movements and take them as speaker cues for speech extraction. Wavoice [4] uses mmWave radar to detect users' vocal vibrations and designs a neural network for multi-modal fusion. UltraSE [6] uses a smartphone to emit ultrasound and capture the ultrasound reflected by the speaker's lips for speech enhancement. *However, video-based approaches require capturing user faces, which suffers from privacy issues. mmWave-based approaches deploy mmWave devices in advance and restrict user movement. Ultrasonic-based approaches require the smartphone stay close to lips, which is inconvenient.*

Previous researches [7], [33]–[36] have explored the possibility of using bone conduction signal for speech enhancement. SEANET [33] and research [35] propose to fuse bone conduction signal and audio signal in time domain. However, fusing multimodal signals in the time domain makes it difficult to leverage the correlation of these signals in the frequency domain. In-Ear-Voice [36] and Wang et al. [34] focuse on leveraging dedicated bone-conduction microphones for speech enhancement, they build a hardware platform and

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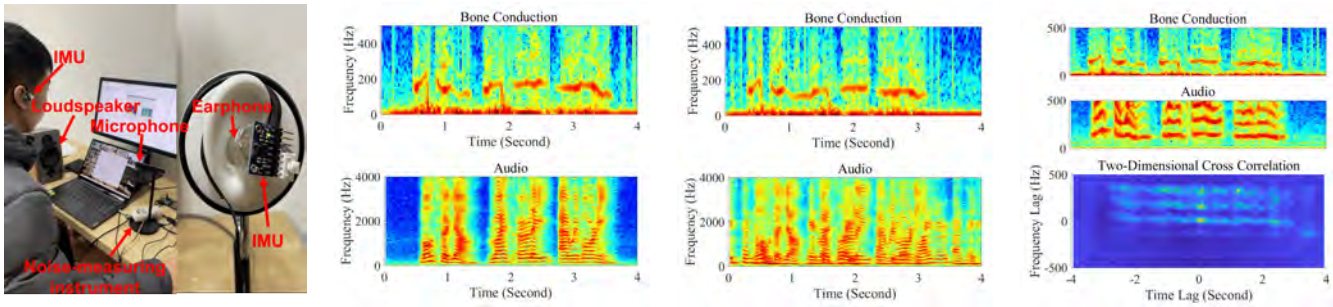

(a) Deployment of empirical study (b) Speech without ambient noise (c) Speech with ambient noise (d) Cross-correlation of audio and bone conduction signal

Fig. 2: Characteristics of bone conduction signal.

design fusing approach. However, dedicated bone-conduction microphones are costly and not integrated into commercial devices. VibVoice [7] is the work closest to ours. It fuses features extracted from bone conducted signals captured by IMU and speech in frequency domain for speech enhancement. However, VibVoice merely concatenates multimodal signals directly and lacks structural guidance to facilitate the capture of correlations between these multimodal signals during the fusion process. These previous studies primarily emphasize data-level fusion, neglecting a comprehensive exploration and exploitation of the fundamental correlations across different modalities. Consequently, there is an inadequate utilization of the bone conduction signal, resulting in unsatisfactory performance under low SNR conditions, as evidenced by the evaluation. Compared to previous researches, *we delve deeply into the correlation between bone conduction signal and audio signal, and design the novel attention-based CTA and FA blocks for effective data fusion. We also propose bone conduction-based noise level metrics and the adaptive model selection approach based on reinforcement learning to innovatively achieve noise reduction acceleration based on noise grading mechanism.* Experimental results show that our approach outperforms audio-based State-of-the-Art (SOTA) method CMGAN [2] and bone conduction-based approachs SEANet [33] and VibVoice [7].

III. Empirical Study and Modeling

In this section, we conduct empirical studies to explore the relationship between the audio signal and bone conduction signal. As shown in Fig. 2(a), considering that commercial earphones do not provide the data interface of the IMU sensor, we attach an MPU6050 IMU sensor to the earphone case to perceive the bone conduction signal. We utilize the microphone to record the user's speech and use a loudspeaker to play ambient noise. The sampling rates of the microphone and IMU are 16kHz and 1kHz, respectively.

Fig. 2(b) shows the bone conduction signal and audio signal when there is no ambient noise. Fig. 2(c) shows multimodal signals when there exists ambient noise. We can find that:

Observation 1: The energy of the perceived bone conduction signal is mainly concentrated in the low-frequency band. The bone conduction signal is not affected by ambient noise. This is because the energy of air-conducted

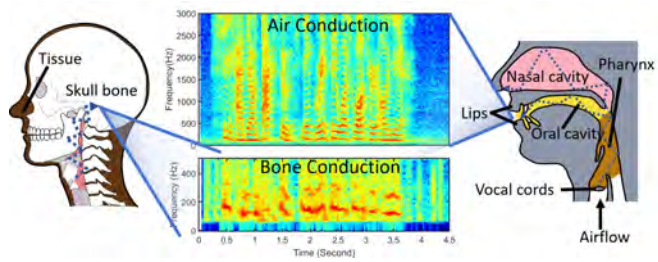

Fig. 3: Air-conducted signal and bone-conducted signal.

sound is not enough to penetrate the skin and be perceived by the IMU sensor. To investigate the relationship between the bone conduction signal and the audio signal, we cross-correlate the multimodal signals. Fig. 2(d) shows the cross-correlation between multimodal signals. We can find that:

Observation 2: The bone conduction signal is highly correlated with the low-frequency component of the audio signal.

According to the above observations, we build a bone conduction model as the basis of multimodal speech enhancement. Specifically, as shown in Fig. 3, when a person speaks, lings expel air flow to cause the vibration of the vocal cord. The vocal cord vibration produces the glottal waveform including fundamental tone and harmonics. Harmonics are generated by standing waves on the vocal cord, and the frequency of harmonics is an integral multiple of the fundamental tone. Therefore, the glottal waveform produced by the vocal cord can be depicted by:

$$v(t) = \sum_{n=1}^{N} r_n(t) cos(2\pi n f_0 t + \phi_n), \tag{1}$$

where $v(t)$ is the glottal waveform, f_0 is the frequency of the fundamental tone, $r_n(t)$, nf_0, and ϕ_n are the amplitude, frequency, and phase offset of the nth harmonic. Then the glottal waveform is modulated by the vocal tract filter (including the pharynx, the oral cavity, and the nasal cavity) to form the air-conducted sound:

$$s_a(t) = v(t) * \delta_a(t) = \sum_{n=1}^{N} a_n(t) r_n(t) cos(2\pi n f_0 t + \phi'_n), \tag{2}$$

where $s_a(t)$ is the air-conducted sound, δ_a is the impulse response of the vocal tract filter, $a_n(t)$ is the vocal tract modulation factor, and $*$ is convolution. Further, the $s_a(t)$

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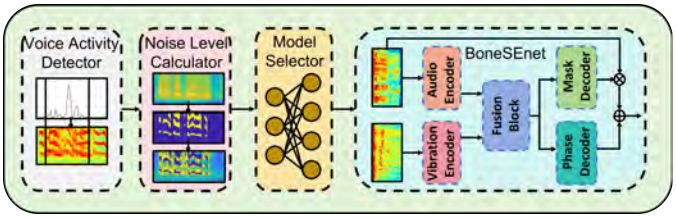

Fig. 4: System overview of BoneSE.

penetrates the vocal tract wall and is modulated by the skull bone and tissue to form the bone-conducted sound:

$$s_b(t) = s_a(t) * \delta_w(t) * \delta_b(t) = \sum_{n=1}^{N} b_n(t) w_n(t) a_n(t) r_n(t) cos(2\pi n f_0 t + \phi_n''), \quad (3)$$

where $\delta_w(t)$ is the impulse response of the vocal tract wall, $\delta_b(t)$ is the impulse response of skull bone and tissue. $b_n(t)$ and $w_n(t)$ are the modulation factors of the vocal tract wall and skull bone. Considering that the empirical study shows that only the fundamental tone and the first harmonic are strong enough to be captured, we can ignore high-frequency harmonics and rewrite Eq. (3) as:

$$s_b(t) \approx b_1(t) w_1(t) a_1(t) r_1(t) cos(2\pi f_0 t + \phi_1'') + b_2(t) w_2(t) a_2(t) r_2(t) cos(4\pi f_0 t + \phi_2''), \quad (4)$$

$b_1(t)w_1(t)$, $b_2(t)w_2(t)$, $a_n(t)$, and $r_n(t)$ are determined by the mechanical properties of the skull bone, tissue, and vocal tract, which are difficult to be described with precise mathematical models. As a result, we are not able to restore air-conducted sound $s_a(t)$ utilizing bone-conducted vibration $s_b(t)$ directly. Fortunately, Eq. (4) shows that the bone-conducted sound shares the same fundamental frequency with the air-conducted sound, which is consistent with the empirical study. Moreover, the air-conducted sound and bone-conducted sound are synchronized in time. On this basis, in this paper, we leverage the time and frequency dependences between multimodal signals for speech enhancement. We introduce the attention mechanism and utilize the bone conduction signal as the attention indicator. We allocate attention to the specific frequencies and time frames of the input audio to separate the user's voice from noisy audio.

IV. System Design

A. System Overview

In this section, we introduce how BoneSE leverages bone conduction signal for speech enhancement. As shown in Fig. 4, BoneSE consists of four components, and takes audio signals and bone conduction signals from a single earphone as inputs. Firstly, we utilize the bone conduction-based voice activity detector to detect the existence of user speech. Specifically, we determine the presence of user voice by detecting whether spectral energy variance (SEV) of bone conduction signal exceeds the threshold. Secondly, we utilize the noise level calculator to obtain the bone conduction-based noise level metrics. Thirdly, we utilize the adaptive model selector to select the suitable denoising model according to the noise level. The adaptive model selector can balance the denoising performance and the time consumption adaptively. Finally, We utilize the selected BoneSEnet to enhance speech. BoneSEnet is an attention-based artificial neural network and utilizes CTA block and FA block to fuse multimodal data.

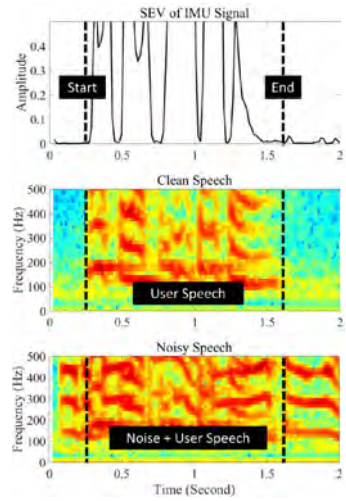

Fig. 5: Voice activity detection.

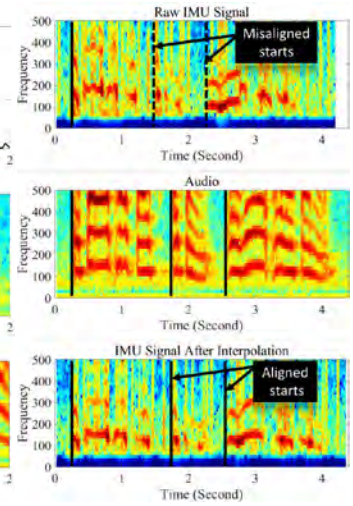

Fig. 6: Interpolation of IMU signal.

B. Bone conduction-based Voice Activity Detector

Considering that speech enhancement with an artificial network is computation-intensive, we propose a voice activity detector (VAD) based on a bone conduction signal to detect the voice activity of the target speaker. And only detected audio signals that contain speeches of the target user will be processed.

Traditional VAD approaches [37], [38] detect voice activity by recognizing the existence of the human voice in audio signals. However, these approaches are unable to distinguish the voice of the target user from external interfering speech. To address the issue, considering that the bone-conducted sound is perceived by contact sensing, which can resist the interference of speeches from non-target users. We propose a target speech activity detector based on the bone conduction energy. Specifically, we first remove the components of body movements in the IMU signals. Body movements are mainly concentrated in the low-frequency bands [39], [40]. Therefore, we apply a high-pass filter with a cut-off frequency of 60 Hz, which is the lower bound of human speech frequency, to filter out human movements. Then, we perform the short-time Fourier transform (STFT) on the IMU signal to obtain spectrogram frames $\{S_i\}$ for voice activity detection. Where $S_i = \{s_i^{f_1}, s_i^{f_2}, s_i^{f_3}, \cdots, s_i^{f_n}\}$ and $s_i^{f_n}$ denotes the spectral energy of frequency bin f_n of the ith frame. We notice that when the user is quiet, the spectral energy distribution of IMU is uniformly low. When the user speaks, the spectral energy distribution becomes uneven. Therefore, we propose to utilize the spectral energy variance (SEV) of IMU as the metric to detect the existence of human speech:

$$SEV(S_i) = \frac{1}{n} \sum_{i=1}^{n} (s_i^{f_n} - \overline{s_i})^2, \quad (5)$$

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where $\overline{s_i}$ is the average energy of all frequencies in frame S_i. Specifically, As shown in Fig. 5, when the SEVs of N_{frame} consecutive frames exceed γ_{start}, we determine that the voice activity starts. Where N_{frame} is the minimum frame number of voice activity and γ_{start} is the start threshold. When the voice activity is in progress, and the SEVs of N_{frame} consecutive frames are lower than the end threshold γ_{end}, we determine that the voice activity is over. After speech activity is detected, we would extract corresponding audio segments for further processing.

C. Adaptive Model Selection Based on Bone Conduction-based Noise Level

To achieve fast processing while ensuring denoising performance, we need to choose the appropriate denoising model according to different noise levels. However, the signal-to-noise ratio is difficult to be obtained in practice because it requires precise energy of both the actual signal and the noise. To address the problem, we propose bone conduction-based noise level metrics to measure the noise level. To accommodate different noise levels, we further propose a model selection approach based on policy gradient to choose the proper model according to the measured noise level.

1) Bone Conduction-based Noise Level Metrics: To measure the noise level accurately, we propose three bone conduction-based noise level metrics, including bone conduction-based signal-to-noise ratio (BSNR), high-energy ratio (BHER), and maximum-energy ratio (BMER). These proposed noise level metrics leverage the intuition that the bone conduction signal is highly correlated with the user's speech, and we can utilize the bone conduction signal as a cue to estimate the noise region and speech region in the audio spectrogram. Before calculation of noise level metrics, we first align the IMU signal with the audio signal. The Short-Time Fourier Transform (STFT) requires uniform sampling. Although we configure the IMU for uniform sampling, we find the actual sampling intervals remained non-uniform in practice. This non-uniformity leads to spectral structure distortion. Furthermore, the bone conduction signal and audio signal fail to align along the time axis. For example, as shown in the top images of Fig. 6, the starts of the bone conduction signal and the audio are unsynchronized because of uneven sampling. We find that such misalignment hinders subsequent feature fusion. To address this issue, we interpolate the IMU signal with the cubic spline interpolation approach to align it with the audio signal. Considering that the IMU sampling rate is 1000 kHz, while the microphone sampling rate is 16 kHz, we set the STFT sliding window for the audio signal to be 16 times that of the IMU signal. This design guarantees that each pixel in the resulting spectrograms corresponds to the same temporal granularity. As shown in the bottom of Fig. 6, the bone conduction signal after interpolation is synchronized with the speech signal.

To obtain noise level metrics, we first binary the spectrogram of bone conduction signal to obtain the mask of speech basic wave. The magnitude of the bone conduction signal varies with different users and different ways of wearing IMU sensors. Therefore, it requires selecting the binarization threshold adaptively. As shown in Fig. 7(a), although the IMU is not affected by external noise, it suffers from the hardware noise due to the limitations of the hardware. The common Otsu's adaptive threshold approach [41] is sensitive to noise and is not able to select an appropriate threshold to filter out hardware noise under such circumstances. To address the issue, we propose a top-down binarization threshold selection method. Specifically, we locate the frequency f_{max} of the bone conduction signal with the maximum energy and take it as the central frequency of the audio fundamental wave. Therefore, the fundamental wave band is $[f_{max} - \delta, f_{max} + \delta]$, where δ is the empirical parameter. Let the binarization threshold start with B, then the number of pixels above B in the fundamental wave band is P_{in}, the number of pixels above B out of the fundamental wave band is P_{out}. We keep decreasing B when the increasing number of pixels in the fundamental wave band is greater than that outside the fundamental wave band:

$$B = B - 1 \times 10^{r-1} \qquad \text{Until } \Delta P_{out} > \Delta P_{in}, \tag{6}$$

where r is the order of magnitude of B. It indicates that the threshold is already between the lowest fundamental energy and the highest noise energy of the bone conduction signal. After binarization with threshold B, we can obtain the binarized spectrogram in Fig. 7(b). We used morphological closure to remove near outliers. When there is more than one connected domain in the same time period, we keep only the largest connected domain. Then we derive the speech mask according to the relationship between harmonic waves and the fundamental wave in speech: $F_{h_i} = i \times F_f$, where F_{h_i} is the frequency of the ith harmonic wave and F_f is the frequency of the fundamental wave. The derived speech mask is shown in Fig. 7(d). The speech mask indicates where the speech energy is concentrated. Fig. 7(e), Fig. 7(f), Fig. 7(g), Fig. 7(h) compare the masked clean speech and masked noisy speech, which indicates that the speech mask can cover the target speech and leave the noise. Leveraging the speech mask, we can obtain the bone conduction-based noise level metrics:

- Bone conduction-based signal-to-noise ratio (BSNR): The ratio of audio energy in the mask to audio energy.
- Bone conduction-based high-energy ratio (BHER): The ratio of the area in mask higher than A to the area outside mask higher than A, where A is the average energy in the mask region.
- Bone conduction-based maximum-energy ratio (BMER): The ratio of the maximum energy in mask to the maximum energy outside the mask.

2) Adaptive Model Selection: To achieve a good balance between noise reduction and processing time, we need to choose the appropriate model according to different noise levels. Inspired by BlockDrop [42], we develop a policy network to select the proper model for speech enhancement and utilize the policy gradient approach for parameter optimization. The model selection can be modeled as a single-step prediction process. As shown in Fig. 8, the policy network takes the proposed BSNR, BHER, and BMER as inputs and outputs

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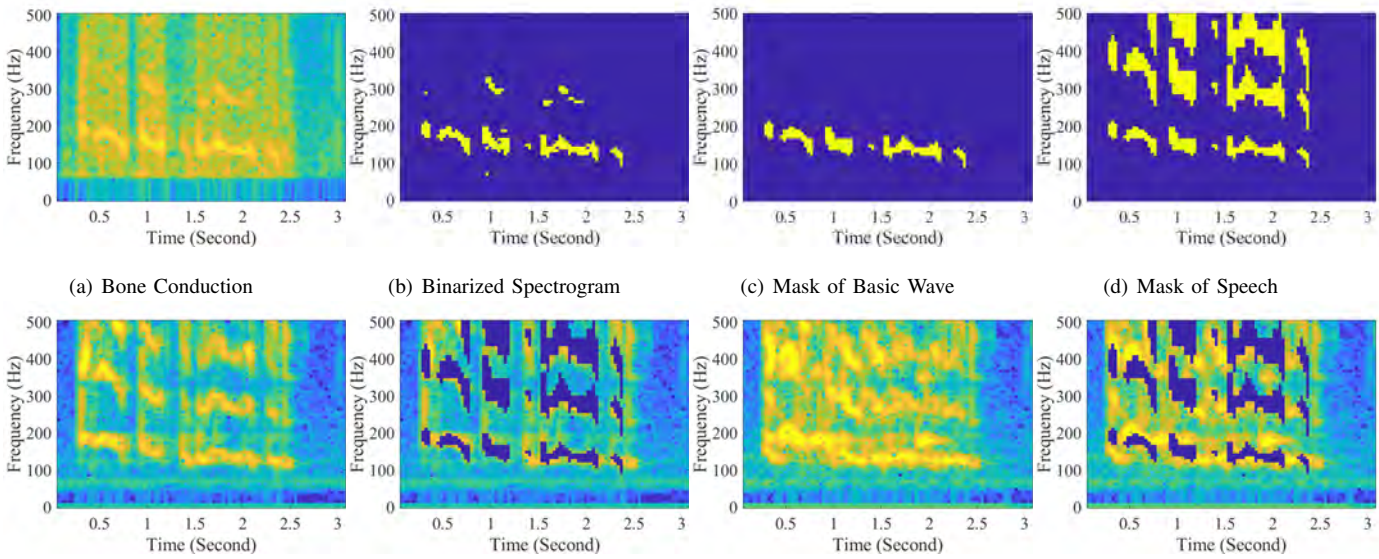

(a) Bone Conduction (b) Binarized Spectrogram (c) Mask of Basic Wave (d) Mask of Speech

(e) Clean Speech (f) Masked Clean Speech (g) Noisy Speech (h) Masked Noisy Speech

Fig. 7: Process of Calculating Bone Conduction-based Noise Level Metrics.

the vector representing the probability of selecting different models.

Specifically, we use f_θ to define a policy network with parameters θ. Therefore, the process of single-step prediction can be represented by $\mathbf{p} = f_\theta(x)$, where x are the obtained bone conduction-based noise level metrics, $\mathbf{p} = (p_1, p_2, ...p_n)$ and p_n represents the likelihood of selecting the ith model. Further, we define the model selection behavior as a categorical distribution D_{cat}, then the model selecting strategy can be obtained by:

$$\pi_\theta(\mathbf{w}|x) = D_{cat}(\mathbf{w}, f_\theta(x)), \tag{7}$$

where $\mathbf{w}$ is the selecting vector, the k-th entry of the vector $w_k \in \{0, 1\}$ and $\sum_k w_k = 1$. $w_k = 1$ indicates that the k-th model is selected. $\pi_\theta(\mathbf{w}|x)$ represents the likelihood of $\mathbf{w}$ given input x. Upon generating the model selection strategy, the model that is selected will be evaluated in the forward pass. For the reward, we set a lower limit for noise reduction performance and encourage short processing time. Therefore, the reward function can be represented by:

$$R(\mathbf{w}) = \begin{cases} -G & SSNR < \lambda_s \\ 1 - \alpha T & SSNR \geq \lambda_s \end{cases} \tag{8}$$

where T is the processing time of the selected model, α is the scale factor, G is positive empirical punishment, and λ_s is the lower limit for SSNR. The goal of training is to maximize the expected reward:

$$L(\theta) = E_{\mathbf{w}\sim\pi_\theta}[R(\mathbf{w})]. \tag{9}$$

To maximize Eq. 9, we utilize the policy gradient approach [43] to train the policy network. By ascending in the direction of the policy parameter gradient, we are able to find the local maximum of $L(\theta)$ by:

$$\theta = \theta + \alpha \nabla_\theta L(\theta). \tag{10}$$

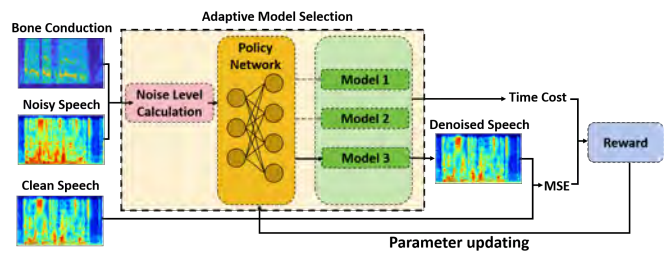

Fig. 8: Training process of the policy network.

Where α is the learning rate. The gradient $\nabla_\theta L(\theta)$ can be computed with REINFORCE algorithm [44] as follows:

$$\nabla_\theta L = E_{\mathbf{w}\sim\pi_\theta}[(R(\mathbf{w}) - R(\hat{\mathbf{w}}))\nabla_\theta log\pi_\theta(\mathbf{w}|x)]. \tag{11}$$

In Eq. 11, we utilize the self-critical baseline [45] $R(\hat{\mathbf{w}})$ to reduce the variance of the gradient estimate. Where $\hat{\mathbf{w}}$ is the strategy with maximum likelihood. In practice, our policy network consists of two fully connected layers and a softmax layer. Therefore, the computational overhead of model selection process is negligible compared to the speech enhancement process.

D. Speech Enhancement

In this section, we introduce how to fuse the bone conduction signal and audio signal for speech enhancement. The intuition is that the bone conduction signal is highly correlated with the low-frequency component of the user's speech. We propose an artificial network called BoneSEnet for speech enhancement, the architecture of BoneSEnet is shown in Fig. 9. BoneSEnet utilizes a two-branch network to extract features from multimodalities, then fuse the features leveraging the correlation between multimodal signals for speech enhancement. BoneSEnet is composed of an audio encoder, bone conduction

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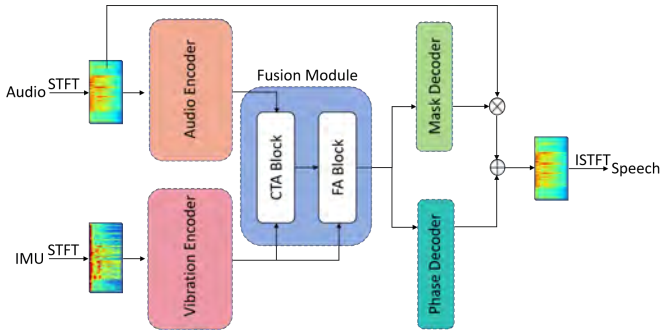

Fig. 9: The Structure of BoneSEnet

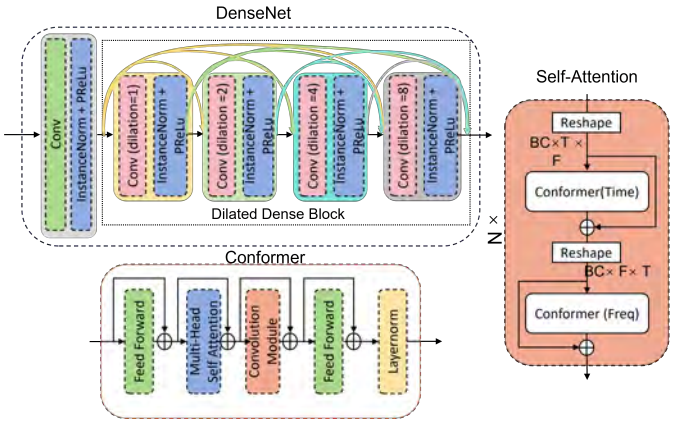

Fig. 10: The structure of encoder.

encoder, fusion module, mask decoder, and phase decoder. The audio encoder and vibration encoder extract and integrate the structural information of the audio signal and the bone conduction signal. The fusion module leverages the CTA block and FA block to allocate attention to the target regions of the audio spectrogram according to the time and frequency correlations between multimodal signals. Finally, the mask decoder decodes the output of the fusion module to obtain the magnitude mask, and the phase decoder rectifies the phase.

1) Audio encoder and vibration encoder.: Previous works [1], [2], [18], [46] show the good performance of speech enhancement in the time-frequency domain. Moreover, empirical studies show that the bone conduction signal has a high similarity to the user's speech in the low-frequency part of the frequency domain. Therefore, we transfer the audio signal and bone conduction signal to the frequency domain to leverage their spectral structure correlation for speech enhancement.

Specifically, the input of the BoneSEnet $\mathbf{I} \in \mathcal{R}^{B\times 3\times T\times F}$ consists of the magnitude, real and imaginary components of the audio spectrogram $\mathbf{U}$. The audio encoder and bone conduction encoder share the same structure, which consists of the densenet and self-attention block. The densenet utilizes a convolution block and a dilated dense block to extract the local features of the spectrogram as shown in Fig. 10. The convolution block utilizes a convolution layer to extend the input spectrogram maps to feature maps with C channels. The dilated dense block is featured by the dense residual connections, and the dilation rate of dilated convolution layers is exponentially increased from 1 to 8. With the increasing dilation rate, the dilated dense block extends the receptive field and helps in spectrogram structure aggregation at different resolutions.

The self-attention block employs conformer blocks to capture the frequency dependency and the time dependency alternatively. As shown in Fig. 10, the conformer block [47] is a convolution-augmented transformer that combines the convolution neural network and transformer [48]. The transformer can capture content-based global features, while convolution exploits local features effectively. The conformer inherits the power of both transformer and convolution and can capture both local and global dependencies of the feature map. The conformer has been proven to be efficient [47] in the field of noise reduction.

2) Channel And Time Attention Block: Most multimodal reduction approaches fuse multimodal features by concatenating them directly, which is a loose way of fusion. However, loose fusion can lead to redundant dependencies between multimodal feature maps and make computing inefficient. To address the issue, we explore and exploit dependencies between multimodal signals. We find that the dependence between speech multimodal signals can be decomposed into time dependence and frequency dependence. We propose the channel and time attention block (CTA) and frequency attention block (FA) to depict the dependence with the attention mechanism to achieve tight fusion.

As shown in Fig. 11(a), the CTA block allocates attention according to the channel and time dependencies. Specifically, we utilize squeeze and excitation [49] to capture the channel dependency. We first employ the global average pooling (GAP) to each channel of the bone conduction feature map to aggregate the global spatial information into a single channel descriptor u_c and obtain the aggregated features:

$$u_c^B = f_{GAP}(x_c^B) = \frac{1}{T\times F}\sum_{i=1}^{T}\sum_{j=1}^{F} x_c^B(i,j), \qquad (12)$$

where $x_c^B \in \mathcal{R}^{T\times F}$ is the bone conduction feature map of channel c. Then we utilize a fully connected layer followed by a ReLu activation to compress C channels into C/r channels for squeeze. Further, we utilize a fully connected layer followed by the sigmoid activation to restore the channel number to C for excitation and obtain the weight of each channel. To allocate attention according to the time dependency, we leverage the synchronization of multimodal signals, i.e., the energy of the multimodal signals concentrates in the same time frame. Specifically, we first employ frequency average pooling (FAP) to obtain the frequency aggregated features of bone conduction signal:

$$g_{t,c}^B = f_{FAP}(x_{t,c}^B) = \frac{1}{F}\sum_{i=1}^{F} x_{t,c}^B(i), \qquad (13)$$

where $x_{t,c}^B \in \mathcal{R}^F$ is the bone conduction feature map of channel c, time t. $g_{t,c}^B$ is the frequency aggregated feature indicating the energy distribution in the time-channel plane. Then we utilize a fully connected layer to obtain statistical

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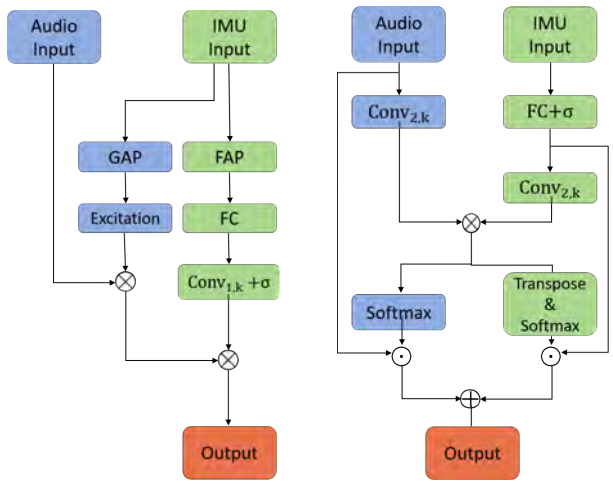

(a) The structure of CTA block (b) The structure of FA block

Fig. 11: Components of Fusion Module.

information of all channels and utilize a 1-dimensional convolution to obtain the weight of time t.

3) Frequency Attention Block: We propose the frequency attention block (FA) to model the frequency dependency. The intuition is that the harmonics of the human voice are strongly correlated to fundamental waves. Considering that the bone conduction signal can capture the low-frequency components of human speech, we can leverage bone conduction signals to restore voice harmonics. Specifically, FA utilizes a two-branch structure for frequency dependency modeling. For the bone conduction branch, we apply a fully connected layer to bone conduction feature maps $x^B \in \mathcal{R}^{C\times T\times F_B}$ to infer voice harmonics and obtain the expanded bone conduction features $e^B \in \mathcal{R}^{C\times T\times F_A}$. e^B has the same frequency dimension as the audio feature map. We utilize the convolution operation to compress C channels into C' channels to reduce the subsequent computation. For the audio branch, we also compress the channel number to C'. Then, we perform matrix multiplication to the compressed multimodal feature maps and obtain the frequency-to-frequency correlation $FFC \in \mathcal{R}^{T\times F_A\times F_A}$:

$$\mathbf{FFC} = CONV_{2,k}(\sigma(FC(\mathbf{x}^B))) \otimes CONV_{2,k}(\mathbf{d}^A). \quad (14)$$

Further, we leverage **FFC** to fuse expanded bone conduction features and audio features. We employ Softmax to normalize the proportion of different frequency components, then perform frequency-wise multiplication to obtain the attention-adjusted features of different modes. Finally, we add these features to get the fusion feature:

$$\mathbf{z} = (SM(\mathbf{FFC}) \odot \mathbf{d}^A) \oplus (SM(\mathbf{FFC'}) \odot \mathbf{e}^B), \quad (15)$$

where $z \in \mathcal{R}^{C\times T\times F_A}$ is the fusion feature, SM is the Softmax operation, **FFC**' is the transpose of **FFC**.

4) Mask Decoder and Phase Decoder: The mask decoder consists of a dilated dense block and two 2 dimension convolution layers. Given the fusion feature, we first employ a sub-pixel convolution layer followed by instance normalization and PReLu activation to expand the feature map to the size of the audio spectrogram. Secondly, we utilize the dilated dense block with dilation rate 1,2,4,8 to aggregate features. Thirdly, we utilize 2 dimension convolution layers followed by PReLu activations and instance normalization to generate the magnitude mask, then we apply the magnitude mask to the input magnitude and obtain the denoised magnitude $\hat{\mathbf{M}}$. The phase decoder has a similar structure as the mask decoder, but the kernel number of the last convolution layer is set to 2 to output the rectified real component and imaginal component $(\hat{\mathbf{U}}_{real}, \hat{\mathbf{U}}_{image})$.

5) Loss Function: We utilize the linear combination of magnitude loss and phase loss as the loss of BoneSEnet:

$$\mathcal{L}_1 = \mathbb{E}_{\hat{\mathbf{M}},\mathbf{M}}(||\mathbf{M} - \hat{\mathbf{M}}||^2), \quad (16)$$

$$\mathcal{L}_2 = \mathbb{E}_{\hat{\mathbf{U}},\mathbf{M}}(||\sqrt{\hat{\mathbf{U}}_{real}^2 + \hat{\mathbf{U}}_{image}^2} - \mathbf{M}||^2), \quad (17)$$

$$\mathcal{L}_3 = \mathbb{E}_{\mathbf{U}_{real},\hat{\mathbf{U}}_{real}}(||\mathbf{U}_{real} - \hat{\mathbf{U}}_{real}||^2) + \mathbb{E}_{\mathbf{U}_{image},\hat{\mathbf{U}}_{image}}(||\mathbf{U}_{image} - \hat{\mathbf{U}}_{image}||^2), \quad (18)$$

$$\mathcal{L}oss = \lambda_1\mathcal{L}_1 + \lambda_2\mathcal{L}_2 + \lambda_3\mathcal{L}_3, \quad (19)$$

where λ_1, λ_2, λ_3 are the weights of corresponding losses. In our experiments, λ_1, λ_2, and λ_3 are set to 0.8, 1.0, and 0.3, respectively.

V. Inplementation and Evaluation

A. Experimental Setup

To evaluate the performance of BoneSE, since the IMU embedded in commercial earphones does not provide an accessible data interface, we mount an MPU6050 IMU sensor on the outer shell of an in-ear earphone using screws. The IMU sensor is interfaced to a Raspberry Pi 4B [50] via I²C for inertial data acquisition, while a USB microphone is connected to the same host to record audio. Both streams are synchronized by referencing a common monotonic system clock. The sampling rates of the microphone and IMU are 16kHz and 1kHz, respectively.

B. Datasets

For user speech collection, we recruit 16 volunteers, including 4 females and 12 males ranging in age from 22 to 29. We utilize the TIMIT Acoustic-Phonetic Continuous Speech Corpus [51] as the phrase list and ask volunteers to practice at least 200 English phrases from TIMIT until they are familiar with them. During data collection, we instruct users to wear an IMU-integrated earphone on their right ear and let volunteers speak phrases at their accustomed pace. Then we collect the IMU signals transmitted by this specific earphone. Samples collected from 8 volunteers are used for training, and samples from the rest volunteers are used for evaluation. We separate the training set from the evaluation set so that categories in the evaluation set are fresh to BoneSEnet.

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	Conv1	Conv2	Conv3	Conv4	Conv5
Num Filters (Large)	32	32	32	32	32
Num Filters (Medium)	16	16	16	16	16
Num Filters (Small)	16	16	16	16	16
Filter Size	3×3	3×3	3×3	3×3	3×3
Stride	(1,2)	(1,1)	(1,1)	(1,1)	(1,1)
Dilation	1	1	2	4	8

TABLE I: Details of Densenet.

	Self-attention		FA block
	N	Dim head	C'
Num (Large)	4	8	16
Num (Medium)	2	8	16
Num (Small)	1	8	8

TABLE II: Details of self-attention block and FA block.

	Conv1	Conv2	Conv3	Conv4	Conv5	Conv6
Num Filters (Mask)	16	16	16	16	16	1
Num Filters (Phase)	16	16	16	16	16	2
Filter Size	3×3	3×3	3×3	3×3	3×3	3×3
Stride	(1,2)	(1,1)	(1,1)	(1,1)	(1,1)	(1,1)
Dilation	1	1	2	4	8	1

TABLE III: Details of the mask decoder and phase decoder.

Model	PESQ	CSIG	CBAK	COVL	SSNR	STOI
/	1.278	2.610	2.429	1.947	5.18	0.800
CMGAN	1.508	2.789	2.658	2.176	6.150	0.798
SEANet	1.315	2.434	2.310	1.893	2.399	0.789
VibVoice	1.878	1.009	3.101	1.244	9.603	0.878
BoneSEnet (Small)	2.110	3.631	3.286	2.918	10.631	0.880
BoneSEnet (Medium)	2.392	3.976	3.507	3.235	11.923	0.903
BoneSEnet (Large)	**2.681**	**4.264**	**3.686**	**3.536**	**12.553**	**0.914**

TABLE IV: Evaluation results on non-speech noise.

For noisy speech synthesis, we adopt two kinds of noise to Synthesize noisy speech datasets. The first kind of noise is randomly chosen from the Nonspeech7k dataset [52]. Nonspeech7k is an audio dataset consisting of human non-speech sounds such as breathing, coughing, and crying. To evaluate the performance of the BoneSE at low SNR, we mix the collected speech samples and audio from Nonspeech7k with a random SNR of -5 to 10 to obtain the synthetic dataset with non-speech noise. The second kind of noise is randomly chosen from the collected speech samples, which can strongly evaluate the ability of BoneSE to distinguish the target speech from external interfering speech. We also mix the collected speech and the noise with a random SNR of -5 to 10 to obtain the synthetic dataset with speech as noise. When SNR is less than 0, the external interfering speech is stronger than the target speech.

C. Network Inplementation

We implement BoneSEnets with varying parameter amounts: small, medium, and large. The details of the Densenet are shown in Table. I. The module number N of the self-attention block, the dim head of the conformer, and the channel number C' are shown in Table. II. The details of the mask decoder and phase decoder are given in Table. III.

Model	PESQ	CSIG	CBAK	COVL	SSNR	STOI
/	1.282	2.960	2.348	2.110	4.498	0.748
CMGAN	1.386	2.627	2.347	2.002	3.781	0.682
SEANet	1.243	2.813	2.135	2.019	1.320	0.739
VibVoice	1.478	1.000	2.642	1.045	6.026	0.793
BoneSEnet (Small)	1.704	3.274	2.839	2.525	7.057	0.820
BoneSEnet (Medium)	1.922	3.614	3.031	2.808	8.311	0.853
BoneSEnet (Large)	**2.413**	**4.124**	**3.412**	**3.317**	**10.305**	**0.904**

TABLE V: Evaluation results on speech noise.

Model	SIG	BAK	OVRL
/	1.6	1.2	1.3
CMGAN	3.2	3.0	3.2
SEANet	3.1	2.9	2.9
VibVoice	4.0	3.6	3.7
BoneSEnet (Small)	4.0	3.8	4.0
BoneSEnet (Medium)	4.3	4.0	4.1
BoneSEnet (Large)	4.7	4.8	4.8

TABLE VI: Subjective Perception Evaluation.

D. Evaluation Metrics

We employ PESQ, CSIG, CBAK, COVL, SSNR, and STOI as the metrics to evaluate the performance of our system. PESQ [53]: Perceptual evaluation of speech quality, ranges from -0.5 to 4.5, a high score indicates the high quality of speech. CSIG [54]: Mean opinion score (MOS) prediction of signal distortion, ranges from 0 to 5, a high score represents low signal distortion. CBAK [54]: MOS prediction of background distortion, ranges from 0 to 5, a high score represents the low background noise interference. COVL [54]: MOS prediction of overall quality of speech, ranges from 0 to 5, a high score indicates high speech quality. SSNR [55]: Segmental signal-to-noise ratio, ranges from 0 to positive infinity, a high score represents high SNR. STOI [56]: Short-time objective intelligibility, ranges from 0 to 1, a high score indicates high intelligibility.

E. Performance Evaluation of BoneSEnet

Evaluation results show that BoneSEnet performs well on two synthetic datasets, and BoneSEnet outperforms the audio-based SOTA approach CMGAN [2], bone-conduction-based approaches SEANet [33] and VibVoice [7] in both objective evaluation and subjective perception evaluation. We first observe and understand different denoising stages of BoneSEnet shown in Fig. 12. In Fig. 12(b), BoneSEnet fuses multimodal signals to find the noise location depicted by the blue shade and denoises the magnitude of the spectrogram by applying the predicted mask. Then, the phase decoder corrects the phase of the spectrogram to obtain the predicted real component Fig. 12(c) and imaginary component Fig. 12(d). Finally, we can obtain the denoised spectrogram shown in Fig. 12(e) by integrating the obtained magnitude and phase. We compare our approach with the audio-based SOTA approach CMGAN and typical multimodal approaches SEANet and VibVoice. To make a fair comparison, we implement these approaches based on the open-source models and train them with our datasets until they converge. Table IV and Table V show the evaluation results on the synthesis dataset with non-speech noise and the synthesis dataset with speech as noise. We

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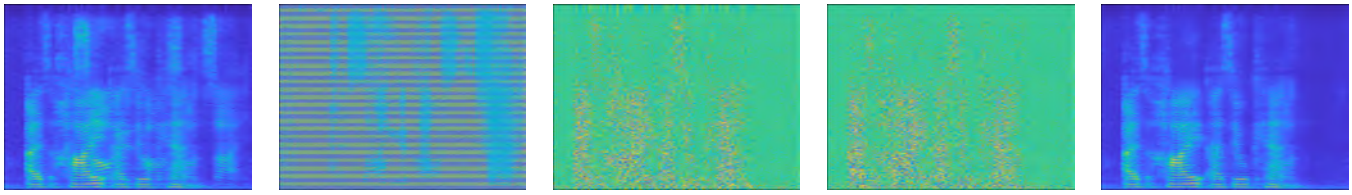


(a) Noisy Audio (b) Predicted Mask (c) Predicted Real Component (d) Predicted Imaginary Component (e) Denoised Audio

Fig. 12: Different stages of noise reduction.

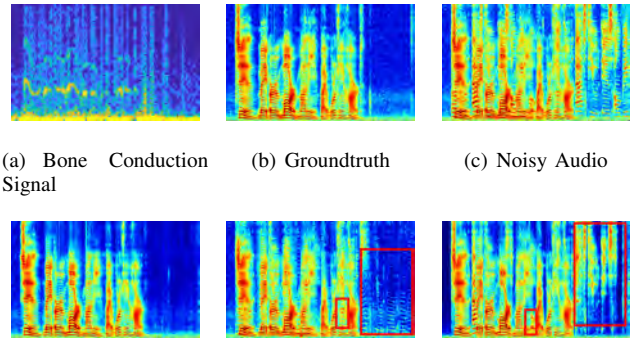
Model	PESQ	CSIG	CBAK	COVL	SSNR	STOI
BoneSEnet (Large)	**2.413**	**4.124**	**3.412**	**3.317**	**10.305**	**0.904**
w/o DenseNet	1.734	3.303	2.870	2.556	7.250	0.837
w/o Self-Attention	1.801	3.406	2.934	2.644	7.620	0.850
w/o FA	1.848	3.599	2.921	2.752	7.586	0.818
w/o CTA	2.359	4.092	3.349	3.273	9.758	0.880

TABLE VII: Ablation Study.

can observe that our approach achieves good performance on both datasets and outperforms comparison approaches on all six evaluation metrics. CMGAN performs badly on the synthesis dataset with speech as noise because the noise is similar to the target speech. CMGAN has no prior information on the target speech and may enhance external interfering speech mistakenly. SEANet is a bone conduction-based speech enhancement approach in the time domain. However, we find that it can not capture the correlation between the multimodal signals and converge very quickly during training. VibVoice fuses multimodal signals in the time-frequency domain by concatenating them directly and lacks of structural guidance to capture correlations between multiple modes. In comparison, BoneSE can capture the correlation between multimodal signals using the CTA and FA blocks, resulting in superior performance even under low SNR conditions. Fig. 13 shows the denoised audio comparison of different approaches. We can observe that VibVoice and CMGAN fail to remove the noise in the red box, because they are not able to capture the relationship between multiple modes. We also conduct a subjective perception evaluation to evaluate the performance of BoneSEnet from human perception perspective. Specifically, we follow the ITU-T P.835 protocol [57], which is a standardized MOS (Mean Opinion Score) test methodology designed for subjective evaluation of speech quality. We select five noisy audios and denoise them with different denoising models. Then 10 volunteers are required to listen to the five audios processed by each model, and comprehensively rate the audio processed by that model on speech quality (SIG), background noise intrusiveness (BAK), and overall quality (OVRL) using a 1–5 scale. Finally, we calculate the average of the volunteers' scores. Results are shown in Table VI, we can observe that BoneSEnet achieves better denoising performance compared with previous works from human perception perspective.

F. Ablation Study

We conduct the ablation study and the results verify the contribution of each block to the BoneSEnet. To conduct the ablation study, we compare the performance of BoneSEnet with the model without DenseNet, the model without Self-Attention Block, the model without FA block, and the model without CTA block. Experimental results shown in Table VII indicate that all the blocks make contributions to BoneSEnet.



(a) Bone Conduction Signal (b) Groundtruth (c) Noisy Audio

(d) Ours (e) VibVoice (f) CMGAN

Fig. 13: Comparision of different approaches.

Model	Time Consumption			Size	Runtime Memory
	GPU	CPU	PHONE		
CMGAN	0.114s	4.69s	35.11s	7.64M	2081M
SEANet	0.009s	0.284s	1.74s	31.9M	109M
VibVoice	0.005s	0.103s	0.81s	11.8M	38.8M
BoneSEnet (Small)	0.0049s	0.093s	0.85s	9.1M	184M
BoneSEnet (Medium)	0.014s	0.117s	1.21s	10.2M	212M
BoneSEnet (Large)	0.022s	0.7s	5.65s	12.3M	519M

TABLE VIII: Time consumption and model size.

G. Performance under different SNR

We evaluate the performance of BoneSEnets under different SNR. Experimental results show that the performance increases with the increase of SNR on six metrics. We make four synthesis datasets with SNR of -5, 0, 5, and 10 using speech as noise. Then we utilize PESQ, CSIG, CBAK, COVL, SSNR, and STOI as metrics to evaluate the performance. The evaluation results are shown in Fig. 14. We can observe that the performance of BoneSEnet increases with the increase of SNR on all six metrics. Under the same SNR, the performance of BoneSEnet (Large) is the best, and the performance of BoneSEnet (Medium) is superior to the performance of BoneSEnet (Small).

H. Time and Power Consumption

We evaluate the time and power consumption of BoneSE. Results show that BoneSEnet can run in real-time on GPU and

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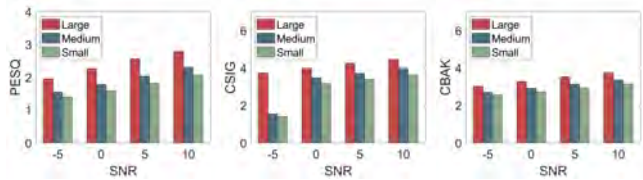
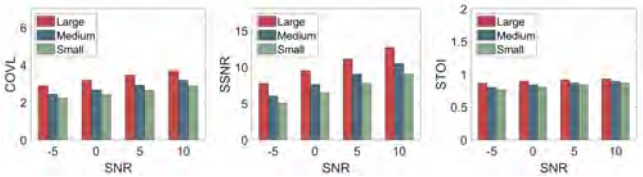


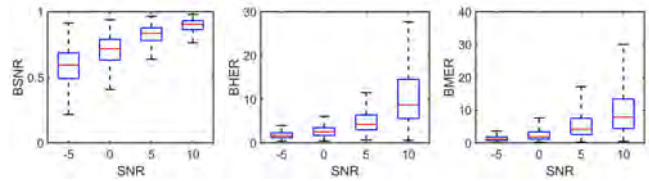
(a) The change tendency of PESQ with SNR (b) The change tendency of CSIG with SNR (c) The change tendency of CBAK with SNR



(d) The change tendency of COVL with SNR (e) The change tendency of SSNR with SNR (f) The change tendency of STOI with SNR

Fig. 14: BoneSEnet performance under different SNR.

the battery capacity of an earphone and phone is enough for BoneSE to work for hours. Specifically, we run BoneSEnet 100 times on different devices to get the average time to process 1s audio and compare it with baseline methods. The GPU model is NVIDIA GeForce RTX 3090, the CPU model is Intel(R) Xeon(R) Silver 4214R CPU @ 2.40GHz, and the smartphone model is OnePlus5. The measurement results are shown in Table. VIII. We can observe that the time consumption of all the models on GPU is less than 0.1s, which indicates that BoneSEnet can run on GPU in real time. In addition, we also measure the runtime memories required for the models to run. The results show that the three bonesenet models required 184M, 212M, and 519M of memory respectively. Compared with the audio-based work CMGAN, BoneSEnet has a faster computing speed and requires less memory during operation. Compared with the most related work Vibvoice, the model size of BoneSEnet is similar to that of Vibvoice, while achieving better denoising performance. This indicates that the proposed BoneSEnet can make better use of the information of input signals for speech enhancement under the condition of similar model parameters. We also evaluate the overall time consumption of the denoising pipeline on the mobile device when λ_S of model selection approach is 5.5. The results show that the time consumption of the denoising pipeline is 4.81s. Among them, the $Noise\ level\ calculation\ \&\ policy\ network$ and $STFT\ \&\ ISTFT$ cost 4.3ms and 46ms respectively, which can be considered negligible compared to the time consumed by the BoneSEnet. For power consumption, the datasheet of MPU6050 sensor [58] shows that the sensor works at 3.7v, 3.6mA when the sampling rate is 1kHz. The battery capacity of an earphone is typically 150mAh, which means that the battery can provide 3.6mA of current for 42 hours at an operating voltage of 3.7v. To assess the energy consumption of the software, we employ the "battery-historian" software to measure the energy consumption of different software modules on OnePlus5 with a battery capacity of 3300mAh.



(a) The change tendency of BSNR with SNR (b) The change tendency of BHER with SNR (c) The change tendency of BMER with SNR

Fig. 15: Evaluation of noise level metrics.

λ_s	PESQ	CSIG	CBAK	COVL	SSNR	STOI	Time(Per second of audio)
6	2.279	3.976	3.310	3.174	9.750	0.893	0.0179s
5.5	2.133	3.352	3.003	2.892	8.483	0.822	0.0143s
5	1.93	3.132	2.77	2.516	8.033	0.767	0.0087s

TABLE IX: Evaluation of adaptive model selection approach.

Specifically, we only run the system processes and measure the basic power consumption of the system first. Then we run $bone\ conduction-based\ voice\ activity\ detection\ module$, $STFT\ \&\ ISTFT$, $policy\ network$, and $BoneSEnet(big)$ separately for one hour to measure their power consumption, and subtract the base power consumption of the system. The results show that the energy consumptions of these four modules in one hour are 78mAh, 175mAh, 151mAh, and 271mAh respectively, and the overall energy consumption is 675mAh. Therefore, the battery of the earphone and smartphone is enough to support the system for hours.

I. Evaluation of bone conduction-based noise level metrics

We evaluate the ability of proposed bone conduction-based noise level metrics in measuring the noise level. Experimental results show that BSNR metrics are positively correlated with the SNR of the signal, which indicates the proposed metrics can reveal the noise level effectively. Specifically, we make four noisy audio datasets with specified signal-to-noise ratios of -5, 0, 5, and 10. Then we calculate the value of BSNR, BHER, and BMER and observe how these values change with the SNR. Experimental results are shown in Fig. 15. We can observe that when SNR increases from -5 to 10, the average of BSNR goes from 0.6 to 0.9, the average of BHER goes from 1.7 to 8.7, and the average of BMER goes from 1.1 to 7.8. This indicates that the proposed metrics are positively correlated with the signal SNR. Therefore, these metrics can reveal the noise level of the signal.

J. Evaluation of Adaptive Model Selection

We evaluate the adaptive model selection approach under different λ_s. Experimental results show that the proposed model selection approach can choose the appropriate model for noise reduction according to the noise level. To evaluate the performance of adaptive model selection, we adopt BoneSEnet (Large) and BoneSEnet (Small) as the possible model options. For training, we make a synthetic speech dataset with a random signal-to-noise ratio of -5 to 10. The scale factor λ is set to 10, the G is set to 1.7, and the λ_s is set to 5, 5.5, and 6.

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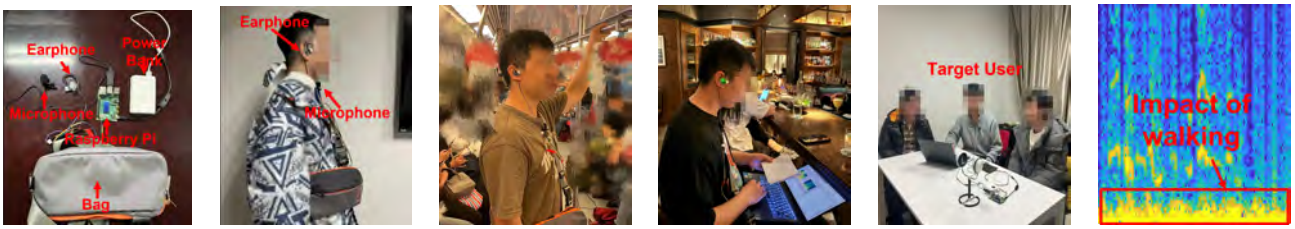

(a) Experimental devices (b) Deployment method (c) Subway scenario (d) Bar scenario (e) Multiple-speaker scenario (f) Bone conduction signal

Fig. 16: Scenarios of case study.

And the training process is conducted with GPU. Experimental results are shown in Table IX. We can observe that the value of metrics increases with the increase of λ_s. This is because the policy network would receive the punishment when the performance is lower than the lower the λ_s. As a result, to achieve good performance, the adaptive model selection approach would select the BoneSEnet (Large) many times, which results in the increasing average time consumption. In this way, we can select a proper denoising model flexibly and balance the performance and time consumption according to actual demand, thereby improving the slow processing speed of always relying on complex parameter-laden denoising networks.

K. End-to-end System Evaluation

We evaluate the end-to-end noise reduction performance of the system, results show that the denoising pipeline of BoneSE can conduct speech enhancement quickly and achieve satisfying performance. Specifically, we construct a new dataset consists of 120 spliced speech samples. Each sample is obtained by first splicing two samples of user speech with a two-second quiet signal, which is collected when the user who wears the earphone keeps quiet. Then we mix spliced speech samples with noise at random SNRs from -5 to 10 to obtain noisy spliced speech samples. For evaluation, we use three pipelines for comparison, complete BoneSE pipeline, the pipeline without Voice Activity Detector (VAD), and the pipeline without model selection. We set the λ_S of model selection to 5.5 and adopt BoneSEnet (Large) and BoneSEnet (Small) as the possible model options. When the VAD does not detect any voice activity, we directly take zero as the value of denoised audio signal. For the pipeline without model selection, we use BoneSEnet(large) for noise reduction. The experimental results are shown in Table X. We can observe that compared with the other two pipelines, the BoneSE pipeline has the shortest processing time and satisfying noise reduction performance. This is because BoneSE pipeline only performs noise reduction on the detected speech segments and can strike a balance between noise reduction performance and processing time. The pipeline without model selection has the longest processing time and the best noise reduction performance, as it always uses BoneSEnet(large) for noise reduction without model selection.

L. Impact of IMU sampling rate, wearing style, untrained sentences, and humming

Pipeline	PESQ	CSIG	CBAK	COVL	SSNR	STOI	Time
BoneSE Pipeline	2.377	3.881	3.403	3.266	9.945	0.839	0.0118s
w/o VAD	2.312	3.763	3.394	3.102	10.134	0.827	0.0141s
w/o Model Selection	2.552	4.331	3.648	3.552	11.015	0.898	0.0193s

TABLE X: Results of End-to-end System Evaluation.

To verify the generalization of the system, we evaluate the impact of IMU sampling rate and wearing style, system performance on untrained sentences and humming. Experimental results show that BoneSE can still achieve good performance when the IMU sampling rate is 500Hz, BoneSE can achieve good performance under different wearing styles, BoneSE can achieve good performance on untrained sentences and humming. Considering that the sampling rates of various IMUs are different, we evaluate the impact of different IMU sampling rates on the system to verify whether the system can generalize to IMUs with low sampling rate. We set the IMU sampling rate to 1kHz, 500Hz, 200Hz respectively and utilize speech noise database to train the BoneSEnet, evaluation results are shown in table XI. We can observe that the system performance decrease with the decreasing of the sampling rate, this is because a high IMU sampling rate can capture a higher frequency of bone conduction signals, allowing the system to learn more correlations between modes. We also evaluate the impact of wearing methods on the system performance. Specifically, we ask users to wear the earphone in three ways to collect data: tight, loose, 90 degrees rotation of earphone. Then we add noises to the collected data and use BoneSEnet(big) for noise reduction. The results in table XI show that all the wearing methods can achieve good noise reduction performance. Interestingly, when wearing loosely, we find that the IMU could better capture the bone conduction signals than wearing earphone tightly. We think this is because the loose fitting makes the IMU more susceptible to bone conduction. To verify if the system works on untrained sentences and non-verbal vocalizations such as humming, we record humming and additional sentences that are not included in the training set, then we add noises for speech enhancement. Denoising results are shown in table XII, we can observe that the system can denoise untrained sentences and humming well. This is because that BoneSE conducts speech enhancement by the correlation between the bone conduction signal and audio signal, which doesn't depend on specific sentences.

M. Case study

We conduct case studies to evaluate the performance of BoneSE in various scenarios: multiple speakers, walking,

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Sampling rate/Wearing style	PESQ	CSIG	CBAK	COVL	SSNR	STOI
1KHz	2.681	4.264	3.686	3.536	12.553	0.914
500Hz	2.425	4.150	3.441	3.336	10.658	0.908
200Hz	2.060	3.814	3.163	2.976	9.414	0.861
Tight	2.396	4.126	3.346	3.305	9.565	0.933
Loose	2.442	4.078	3.399	3.307	9.933	0.932
90 degrees rotation	2.477	4.145	3.421	3.359	10.001	0.938

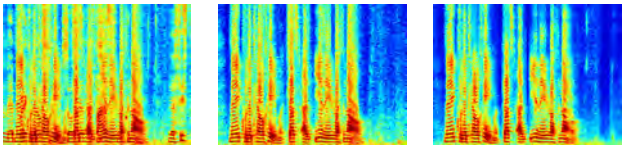
TABLE XI: Impact of IMU sampling rate and wearing style.

Type	PESQ	CSIG	CBAK	COVL	SSNR	STOI
Untrained sentences	2.181	3.895	3.236	3.085	9.268	0.910
Humming	2.012	3.799	3.222	2.945	10.655	0.703

TABLE XII: Performance on untrained sentences and humming

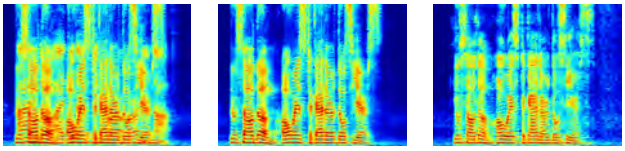
Scenario	PESQ	CSIG	CBAK	COVL	SSNR	STOI
Walking	2.596	4.211	3.409	3.449	9.982	0.918
Running	1.396	3.172	2.638	2.317	6.257	0.737
Boxing	1.548	3.322	2.761	2.472	6.906	0.781

TABLE XIII: Evaluation results of different scenarios.



(a) Noisy audio (b) Denoised audio (c) Reference audio

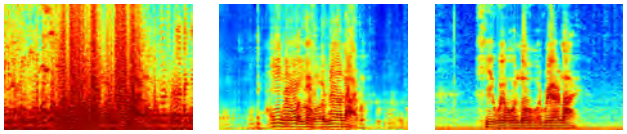
Fig. 17: Audio sample of walking scenario.



(a) Noisy audio (b) Denoised audio (c) Reference audio

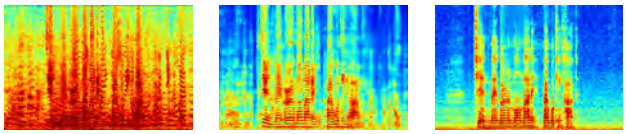
Fig. 18: Audio sample of multiple-speaker scenario.

running, boxing, subway, and bar as shown in Fig. 16. Experimental results demonstrate that BoneSE achieves favorable performance across most scenarios. However, intense physical activity tends to compromise its noise reduction efficacy. Specifically, as shown in Fig. 16(a) and Fig. 16(b), to enable the user to carry the devices while moving, we employ a power bank as the power supply and put the required devices in a bag. For the walking, running, boxing scenarios, we let the user speak while walking and collect 60 pieces of audio data from three users. The collected audio samples are mixed with noise to synthesize noisy audio for evaluation. Experimental results are shown in Fig. 17 and Table XIII. The BoneSEnet (Large) achieves good performance in the running senario. In the scenarios of running and boxing, the system scores low in six indexes. This is because the high-pass filtering of BoneSE can effectively eliminate the walking-induced interference illustrated in Fig. 16(f). However, intense exercise can generate high-frequency motion noise in the IMU, and this noise is difficult to be filtered out by high-pass filtering. As a result, the performance declines. Nevertheless, it should be noted that speech demands are usually low during intense exercise, which means that intense exercise is not a major application scenario for noise reduction algorithms.



(a) Noisy audio (b) Denoised audio (c) Reference audio

Fig. 19: Audio sample of subway scenario.



(a) Noisy audio (b) Denoised audio (c) Reference audio

Fig. 20: Audio sample of bar scenario.

The denoising results of the multiple-speaker scenario, subway scenario, and bar scenario are shown in Fig. 18, Fig. 19, and Fig. 20, respectively. Considering that the ground truth is not available in these scenarios, we use the audio recorded in the quiet environment as the reference audio to visually compare the noise reduction results. In Fig. 18(b), we can observe that BoneSE successfully separates the target speech from the mixed speech of multiple people. In the very noisy subway and bar, as shown in Fig. 19(a) and Fig. 20(a), we can notice that the speeches of the target user are buried in the ambient noise. The denoised audios are shown in Fig. 19(b) and Fig. 20(b). The reference audios are shown in Fig. 19(c) and Fig. 20(c). By comparison, we can find that the speeches of the target user are extracted successfully, which reveals that the proposed approach can enhance speech under the condition of low SNR and proves the superiority of our work.

N. Discussion

A bold idea is that whether we can reconstruct speech directly using bone conduction signals. To verify this idea, we attempt to reconstruct users' speech using only bone-conducted signals, but the performance is unsatisfying. We attribute this to the excessive loss of the original speech information in bone-conducted signals. The user's bones and skin can be regarded as a filter, whose parameters vary substantially across individual users. The information of original speech would get lost during the filtering process and it is extremely difficult to reverse this process with only bone-conducted signals for speech restoration. Therefore, in this paper, we just utilize bone-conducted signals as a side-channel to assist in speech noise reduction.

VI. Conclusion

In this paper, we propose BoneSE, a bone conduction-assisted speech enhancement approach. We delve deeply into the correlation between air-conducted sound and bone-conducted sound. Compared with previous multimodal speech

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enhancement approaches, we innovatively leverage the correlation between multiple modalities to guide attention allocation for data fusion. We propose a bone conduction-based noise level grading mechanism and an adaptive model selection approach to speed up the denoising process by choosing appropriate models for different noise levels. We implement the BoneSE system and evaluate its performance on our datasets. Experiment results show that BoneSE outperforms the state-of-the-art approaches on two datasets and can enhance speech in low SNR scenarios such as subway and bar.

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